

Ch21a: Quantum Chemistry

Alessio Amaolo

December 2018

Introduction

The latter half of Ch21a explores some applications to chemistry. The purpose of this document is to attempt to organise and explain my notes from that section of the course.

It is assumed that the reader already learned the initial part of the course, up to basic quantum mechanics of many particle systems in three dimensions.

Textbook: Griffiths, *Introduction to Quantum Mechanics* (Second Edition).

Contents

1	Many Electron Atoms	2
1.1	Theory	2
1.2	The Helium Atom	3
2	Molecules	3
2.1	Theory	3
2.2	Molecular Orbital Theory	4
2.2.1	Heteronuclear Diatomics	6
2.3	Hybridization	6
2.4	Conjugated Systems	7
2.4.1	π Molecular Orbital Theory	7
3	The Variational Theorem	8
4	Perturbation Theory	10
4.1	Theory	10
4.2	Application: Molecule in an Electric Field	10
4.3	Application: Intermolecular Interactions	11
4.3.1	Theory	11
4.3.2	The Vanderwaals Force	12

1 Many Electron Atoms

1.1 Theory

Suppose we have a nucleus with Z protons, and thus with charge Ze , along with Z electrons in its orbitals. Writing the exact Hamiltonian for the electrons in this system is not difficult.

$$\hat{H}_{el} = \hat{T}_{el} + \hat{V}_{ne} + \hat{V}_{ee}$$

Where $\hat{T}_{el}$ is the kinetic energy of the electrons in the system, $\hat{V}_{ne}$ is the potential energy resulting from interactions between the nucleus and the electrons, and $\hat{V}_{ee}$ is the potential energy resulting from the repulsions between electrons.

If these potentials are simply Coulombic, we get that

$$\hat{H}_{el} = \underbrace{\sum_i \frac{-\hbar^2}{2m} \nabla_i^2}_{\text{Kinetic Energy}} - \underbrace{\sum_i \frac{Ze^2}{r_i 4\pi\epsilon_0}}_{\text{Nuclear Attraction}} + \underbrace{\sum_{i>j} \frac{e^2}{4\pi\epsilon_0 |r_i - r_j|}}_{\text{Electronic Repulsion}}$$

We sum over $i > j$ so that we count every pair of electrons but we do not double count their interactions, and we do not count an electron interacting with itself.

It is fairly difficult to solve the Schrodinger equation with this Hamiltonian, so we resort to approximations. First, it would be convenient if the potential of an electron i were a function only of its own position rather than the position of other electrons. We assume that this is true, and that electron i feels some effective *average* potential related only to its position. While some potential function does exist to make this exact, such a function would be impossible to find. Thus, we make do with some approximate function that we will see is radial in nature. Now that Hamiltonian for an electron is only independent of other electrons, the total Hamiltonian is the sum of the Hamiltonians of each electron. We write that simply $\hat{H} = \sum_i \hat{h}(i)$.

To simplify things even further, consider (at least intuitively) that electrons are generally close to the nucleus, and are an “average” distance from the nucleus. The closer an electron is to the nucleus, the more it will be repelled outwards by other electrons. Thus, it is reasonable to assume that the electronic repulsion potential term is radial. Hence,

$$\hat{h} = -\frac{\hbar^2}{2m} \nabla_i^2 + \hat{V}_{ne}(r_i) + \hat{V}_{eff}(r_i)$$

for a spherical $\hat{V}_{eff}$.

We already know the solutions to the Schrodinger equation for this Hamiltonian, since it looks identical to the hydrogen atom.

$$\begin{aligned} \hat{h}\phi_i(\vec{r}) &= \epsilon_i \phi_i(\vec{r}) \\ \phi_i(\vec{r}) &= R_{nl}(\vec{r}) Y_{lm}(\theta, \varphi) \end{aligned}$$

These are the the same orbitals that we are familiar with, and are identical to the hydrogen orbitals but with a different potential and thus different energies ϵ_i . Overall, $E = \sum \epsilon_i$. Note that

the Hamiltonian commutes with both $\hat{S}_z$ and $\hat{S}^2$, so eigenfunctions of the Hamiltonian are eigenfunctions of spin. Since an electron has two spin states, there exists two degenerate eigenfunctions.

$$\begin{aligned}\hat{h}|\phi_i \uparrow\rangle &= \epsilon_i |\phi_i \uparrow\rangle & \text{for } \phi_i^\uparrow(r) \text{ or } \phi_i^\alpha(r) \\ \hat{h}|\phi_i \downarrow\rangle &= \epsilon_i |\phi_i \downarrow\rangle & \text{for } \phi_i^\downarrow(r) \text{ or } \phi_i^\beta(r)\end{aligned}$$

In an atom, this degeneracy is lifted due to spin orbit coupling.

Further Reading: Drago, *Physical Methods for Chemists* (Second Edition), Chapter 10: “The Electronic Structure and Spectra of Transition Metal Ions”

Determining a $\hat{V}_{eff}$ such that the problem remains easy to solve but gives meaningful results is a difficult problem and one that will not be covered in this class.

Further Reading:

1.2 The Helium Atom

This is covered completely in the textbook in section 5.2.1. Before proceeding, become familiar with the structure of atoms with multiple protons, the Slater determinant (and the Antisymmetrizer), paying particular importance to why electrons in the same orbital must have opposite spin.

2 Molecules

2.1 Theory

We start with the same principle that we started with to examine many electron atoms: by writing out the exact Hamiltonian for the electrons. For a molecule with N electrons and M stationary nuclei,

$$\hat{H}_{el} = \underbrace{\sum_{i=1}^N \frac{-\hbar^2}{2m} \nabla_i^2}_{\text{Electron Kinetic Energy}} - \underbrace{\sum_{i=1}^N \sum_{A=1}^M \frac{Z_A e}{4\pi\epsilon_0 |r_i - R_A|}}_{\text{Nuclei Electron Attraction}} + \underbrace{\sum_{i>j} \frac{e^2}{4\pi\epsilon_0 |r_i - r_j|}}_{\text{Electron-Electron Repulsion}}$$

The notation is a little different than before (we’ve summed from 1 to N and 1 to M to keep track of electrons and nuclei), but we haven’t changed anything. The second term is just the Coulombic attraction between an electron and each nuclei A at position R_A and charge Z_A . The other terms remain identical to the many electron atom.

Then, we write the electron-electron repulsion potential term as an effective potential as a function of position of each electron. Remember, this function may be complicated, but this function exists. Thus, we have not yet made any approximations.

Overall, the Hamiltonian $\hat{h}$ for a single electron is

$$\hat{h}(\vec{r}) = -\frac{\hbar^2}{2m} \nabla^2 + \hat{V}_{ne}(\vec{r}) + \hat{V}_{eff}(\vec{r})$$

With

$$\hat{h}\phi_i^\sigma = \epsilon_i\phi_i^\sigma \quad \text{for } \sigma = \alpha, \beta$$

We call ϕ_i^σ molecular orbitals.

2.2 Molecular Orbital Theory

Now that we've established the basic theory, we can start making approximations. We now assume that $\hat{H} = \sum \hat{h}(\vec{r}_i)$, and that the overall wavefunction $\psi(r_1, r_2, \dots, r_n)$ is the product of molecular orbitals, adjusted for antisymmetry. Thus,

$$\psi = \mathcal{A}\phi_1^\alpha(r_1)\phi_1^\beta(r_2)\phi_2^\alpha(r_3)\dots$$

We don't know yet what molecular orbitals look like. Suppose we have the molecule H_2 . Since atomic orbitals are a complete set of functions, we can write a molecular orbital as linear combination of atomic orbitals on each of the hydrogen atoms A and B.

$$\phi = c_{1sA}1s_A + c_{2sA}2s_A + \dots + c_{1sB}1s_B + \dots$$

The Atomic Orbital expansion is complete (so this is not an approximation) but it is not orthonormal set. Now, we assume that we only care about the lower energy molecular orbitals (so the contribution from the other ones will be negligible), so

$$\phi = c_{1sA}1s_A + c_{1sB}1s_B$$

Since we are just using a single s orbital in each atom, let $c_{1sA} = c_A$ and $1s_A = s_A$ (likewise for B). The next assumption is that ϕ is an eigenfunction of $\hat{h}$. So we have

$$\hat{h}[c_A|s_A\rangle + c_B|s_B\rangle] = \epsilon[c_A|s_A\rangle + c_B|s_B\rangle]$$

.

Multiplying by $\langle s_A|$ or $\langle s_B|$, we get two different expressions.

$$\begin{aligned}\langle s_A|\hat{h}|c_A s_A + c_B s_B\rangle &= \epsilon \langle s_A|c_A s_A + c_B s_B\rangle \\ \langle s_B|\hat{h}|c_A s_A + c_B s_B\rangle &= \epsilon \langle s_B|c_A s_A + c_B s_B\rangle\end{aligned}$$

We can think of $\langle s_A|\hat{h}|s_B\rangle$, informally, as the energy of overlap between A and B. We let $h_{AA} = \langle s_A|\hat{h}|s_A\rangle$, $h_{AB} = \langle s_A|\hat{h}|s_B\rangle$, $S_{AA} = \langle s_A|s_A\rangle$, $S_{AB} = \langle s_A|s_B\rangle$, and so on. We can now represent these two equations as

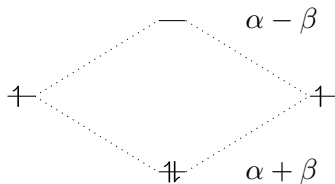
$$\begin{bmatrix} h_{AA} & h_{AB} \\ h_{BA} & h_{BB} \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = \epsilon \begin{bmatrix} S_{AA} & S_{AB} \\ S_{BA} & S_{BB} \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix}$$

Since s_A and s_B are normalized, we know $\langle s_A|s_A\rangle = \langle s_B|s_B\rangle = 1$. For the purposes of simplifying the problem, we assume s_A and s_B are orthogonal, so $s_{AB} = s_{BA} = 0$.

For identical atoms (just like H_2), we let $h_{AA} = h_{BB} = \alpha$ and $h_{AB} = h_{BA} = \beta$. Overall,

$$\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = \epsilon \begin{bmatrix} c_A \\ c_B \end{bmatrix}$$

which has eigenvalues $\epsilon = \alpha \pm \beta$. By convention, the energy of overlap of orbitals is negative, so $\beta < 0$. This gives rise to the standard MO theory diagram for H_2 .



The eigenfunctions are

$$\begin{aligned}\phi_+ &= \frac{1}{\sqrt{2}}(s_A + s_B) \quad \text{for } \epsilon = \alpha + \beta \\ \phi_- &= \frac{1}{\sqrt{2}}(s_A - s_B) \quad \text{for } \epsilon = \alpha - \beta\end{aligned}$$

These are the bonding and antibonding molecular orbitals, respectively.

In general, we have a large matrix with an arbitrary number of molecular orbitals, all linear combinations of atomic orbitals. This yields the typical MO theory diagram with many energies that combines any two types of orbitals.

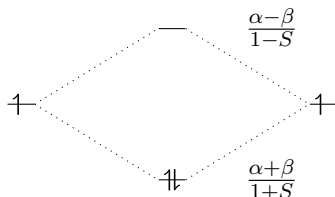
The lines we draw in a standard MO diagram assume that molecular orbitals are made of only two atomic orbitals. The reality is that every molecular orbital is a linear combination of every atomic orbital. However, we assume that the biggest contribution is from the two atomic orbitals nearest in energy.

In general, the matrix equation we must solve for an n electron system to get the energies and eigenfunctions is

$$\begin{bmatrix} h_{11} & \dots & h_{1n} \\ \vdots & \ddots & \vdots \\ h_{n1} & \dots & h_{nn} \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \epsilon \begin{bmatrix} S_{11} & \dots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{n1} & \dots & S_{nn} \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

where, assuming orthogonality, the matrix of S_{ij} is the identity matrix.

What happens if we do not assume orthogonality? In the two by two case, let $S_{AB} = S_{BA} = S$. Then, the eigenvalues are $\epsilon_{\pm} = \frac{\alpha \pm \beta}{1 \pm S}$. The molecular orbital diagram for H_2 is



Since $0 \leq S \leq 1$, we can conclude that the antibonding orbital is shifted up in energy and the bonding orbital is shifted down in energy as compared to the case where we assumed orthogonality. This is why in general chemistry, we say “antibonding orbital is more antibonding than bonding is bonding.” For the rest of these notes, however, we assume orthogonality to simplify the finding of eigenvalues and eigenfunctions.

2.2.1 Heteronuclear Diatomics

Since we have assumed $h_{AA} = h_{BB} = \alpha$, we have assumed that both atoms in the diatomic molecule are identical. Now, we attempt to do the MO theory for the HeH molecule. Again, we assume that the molecular orbitals are composed of linear combinations of only the lowest energy atomic orbitals.

$$\phi = c_A s_A^H + c_B s_B^{He}$$

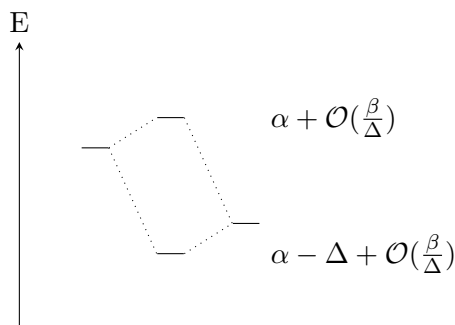
We write $h_{AA} = \alpha$ and $h_{BB} = \alpha - \Delta$, where atom A is Hydrogen and B is Helium. Thus,

$$\begin{bmatrix} \alpha & \beta \\ \beta & \alpha - \Delta \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = \epsilon \begin{bmatrix} c_A \\ c_B \end{bmatrix}$$

which has eigenvalues $\epsilon_{\pm} = \frac{1}{2}[2\alpha - \Delta \pm \sqrt{4\beta^2 + \Delta^2}]$. This makes sense, since as $\Delta \rightarrow 0$, $\epsilon_{\pm} \rightarrow \alpha \pm \beta$, the values we got before. If $\Delta \gg \beta$ (as would be the case for two atoms of very different electronegativities), then $\epsilon_{\pm} = \frac{1}{2}[2\alpha - \Delta \pm \Delta + \mathcal{O}(\frac{\beta}{\Delta})]$, which means that

$$\begin{aligned} \epsilon_+ &= \alpha + \mathcal{O}\left(\frac{\beta}{\Delta}\right) \\ \epsilon_- &= \alpha - \Delta + \mathcal{O}\left(\frac{\beta}{\Delta}\right) \end{aligned}$$

This makes sense. If two atoms are very different in electronegativity, the Molecular Orbital diagram will have very small raising or lowering of energies (on the order of $\mathcal{O}(\frac{\beta}{\Delta})$), a fact learned (qualitatively) in general chemistry.



2.3 Hybridization

Suppose that we are trying to solve for the wavefunction of water. If we had a computer, we could expand it in terms of the atomic orbitals of both hydrogen atoms and the oxygen atom. But by hand, we can choose linear combinations of a selection atomic orbitals to be the basis for the wavefunction. We write

$$\begin{aligned} sp_1^3 &\text{ as } 2s + 2p_x + 2p_y + 2p_z \\ sp_2^3 &\text{ as } 2s - 2p_x - 2p_y + 2p_z \\ sp_3^3 &\text{ as } 2s + 2p_x - 2p_y - 2p_z \\ sp_4^3 &\text{ as } 2s + 2p_x - 2p_y + 2p_z \end{aligned}$$

This particular set is convenient because it is a basis that points to the four corners of a tetrahedron. The matrix that we must solve the eigenvalues for is then a 6×6 matrix with columns representing the interactions between each pair of the four hybridized orbitals and two hydrogen s orbitals.

2.4 Conjugated Systems

We turn our attention to conjugated systems. For the cases analysed here, they are planar molecules with alternating single and double bonds. We will develop a method to determine the energies of these systems.

2.4.1 π Molecular Orbital Theory

We examine the case of C_4H_6 , or 1,3-Butadiene. We expand the π molecular orbital system as a linear combination of the p_z atomic orbitals of the constituent carbons.

$$\pi = c_1 p_1 + c_2 p_2 + c_3 p_3 + c_4 p_4$$

Making the same approximation as before, $\langle p_i | \hat{h} | p_i \rangle = \alpha$ and $\langle p_i | \hat{h} | p_j \rangle = \beta$ if i, j are consecutive, or 0 otherwise. Furthermore, $\langle p_i | p_i \rangle = 1$ and $\langle p_i | p_j \rangle = 0$. The matrix equation is

$$\begin{bmatrix} \alpha & \beta & 0 & 0 \\ \beta & \alpha & \beta & 0 \\ 0 & \beta & \alpha & \beta \\ 0 & 0 & \beta & \alpha \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \epsilon \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

The eigenvalues and eigenfunctions (excluding coefficients that normalise them) are

$$\begin{aligned} \epsilon &= \alpha + 1.618\beta & \text{for } \phi_1 &= p_1 + p_2 + p_3 + p_4 \\ \epsilon &= \alpha + 0.618\beta & \text{for } \phi_2 &= p_1 + p_2 - p_3 - p_4 \\ \epsilon &= \alpha - 0.618\beta & \text{for } \phi_3 &= p_1 - p_2 - p_3 + p_4 \\ \epsilon &= \alpha - 1.618\beta & \text{for } \phi_4 &= p_1 - p_2 + p_3 - p_4 \end{aligned}$$

These are the four different eigenstates. The lowest energy one has energy $\alpha + 1.618\beta$ (remember $\beta < 0$) and represents ϕ_1 , i.e. all the p orbitals being positive on the same side of the molecule and thus the electrons being shared throughout this system. The highest energy one has energy $\alpha - 1.618\beta$ and has an eigenstate with a nodal plane between each atom.

Since each carbon supplies a single electron and there are four carbons, the two lowest molecular orbitals are occupied. Thus, the total energy is $E = 4\alpha + 4.47\beta$.

Suppose we were to attempt to make the molecule non-planar such that the p_z orbitals of carbon three was perpendicular to the same orbital of carbon two. Then $\langle p_2 | \hat{h} | p_3 \rangle = 0$, so the matrix equation is

$$\begin{bmatrix} \alpha & \beta & 0 & 0 \\ \beta & \alpha & 0 & 0 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & \beta & \alpha \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \epsilon \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

This matrix has eigenvalues $\epsilon = \alpha - \beta, \alpha - \beta, \alpha + \beta, \alpha + \beta$. With four electrons, the total energy is $E = 4\alpha + 4\beta > 4\alpha + 4.47\beta$, so this system is more unstable than planar 1,3-Butadiene.

Finally, we can compute the energies of C_4H_4 , or cyclic butadiene. The matrix is

$$\begin{bmatrix} \alpha & \beta & 0 & \beta \\ \beta & \alpha & \beta & 0 \\ 0 & \beta & \alpha & \beta \\ \beta & 0 & \beta & \alpha \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \epsilon \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

The eigenvalues are $\epsilon = \alpha + 2\beta, \alpha, \alpha, \alpha - 2\beta$. These can be drawn at each vertex of the cyclic molecule, and this the Frost circle seen in general chemistry. The total energy is $E = 4\alpha + 2\beta$, which is higher than the linear structure. Thus, the ring conjugated system is destabilizing, and we call this *antiaromaticity*.

Doing the same computation for Benzene, the eigenvalues are $\epsilon = \alpha + 2\beta, \alpha + \beta, \alpha + \beta, \alpha - \beta, \alpha - \beta, \alpha - 2\beta$, which also form a frost circle. In this case, the total energy is $6\alpha + 8\beta$, which is lower than the energy of linear hexatriene with energy $6\alpha + 6.69\beta$. Thus, benzene is stabilised by being in cyclic form, and we call this *aromaticity*.

There is a not-necessarily deep but interesting connection here to another problem in physics. The matrix problem for cyclic conjugated systems looks identical to the one for a translationally invariant system of harmonic oscillators. The eigenfunctions that describe this system are simply the normal mode solutions to this classical problem.

3 The Variational Theorem

Suppose we have two approximate solutions to $H\phi = E\phi$. Which one is better? One thing we could try is finding the energy of each one, and choosing the one that is lower.

Variational Theorem. for any approximate solution ϕ to the time independent Schrodinger equation,

$$\frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} \geq E_0$$

where E_0 is the ground state energy of the true solution to the same equation. If equality holds, then ϕ must be one of the degenerate eigenstates of the ground state of $\hat{H}$.

Proof. Let ψ_i be the eigenstates of $\hat{H}$. Since they are complete, we can write $\phi = \sum_i c_i \psi_i$. Without loss of generality, ϕ is normalized, so $\sum_i |c_i|^2 = 1$. $E = \langle \phi | \hat{H} | \phi \rangle = \sum_i \sum_j c_i^* c_j \langle \psi_i | \hat{H} | \psi_j \rangle$. We know that $\hat{H} |\psi_j\rangle = E_j |\psi_j\rangle$ and ψ_i is orthogonal to ψ_j hence the sum is non-zero when $j = i$. Hence $E = \sum_i c_i^* c_i E_i = \sum_i |c_i|^2 E_i \geq E_0$.

We will not prove the statement regarding equality of the guess and a degenerate eigenstate.

This suggests a natural computational procedure to approximate the ground state of any Hamiltonian. We choose a function $\psi(c_i)$ based on some number of parameters c_i . Then, we compute the expectation value of the Hamiltonian (the energy) and we use our favourite method to find the global minimum of a function with the respect to these parameters.

Example (Harmonic Oscillator). Suppose we want to find the ground state energy of the one dimensional harmonic oscillator. This is a convenient example, as it is one that we've solved

exactly and will illustrate the use of this method. As before, $\hat{H} = -\frac{\hbar}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2}mw^2x^2$. We choose a test function $\psi(x) = Ae^{-bx^2}$, with b as our parameter. Normalizing it, $A = (\frac{2b}{\pi})^{1/4}$.

$$E = \langle \psi | \hat{H} | \psi \rangle = \frac{\hbar^2 b}{2m} + \frac{mw^2}{2b}$$

. We can find the minimum of this function by simply computing

$$\frac{\partial E}{\partial b} = 0 \implies b = \frac{mw}{2\hbar} \implies E = \frac{1}{2}\hbar w$$

This is the exact energy, which tells us that the function we've chosen is the true wavefunction for the harmonic oscillator.

Why did we get the same answer as the analytic solution? We cheated, because we chose a function that is of the same form as the analytic solution, so it can be tuned to the analytic solution.

Example (Molecular Orbital Theory). We will now use the variational theorem to show an independent way to give rise to the same method to calculate the wavefunction of a diatomic molecule. As before, we assume orthogonality of two distinct states. Rather than name specific orbitals, however, let the bonding orbital of the first atom be η_1 and of the second atom be η_2 (more precisely, the orbital we choose to make up our molecular orbital, as the exact solution would require every atomic orbital to be used). Then, $\phi = c_1\eta_1 + c_2\eta_2$. Using the variational theorem,

$$\frac{\langle \phi | \hat{h} | \phi \rangle}{\langle \phi | \phi \rangle} = \frac{\langle c_1\eta_1 + c_2\eta_2 | \hat{h} | c_1\eta_1 + c_2\eta_2 \rangle}{\langle c_1\eta_1 + c_2\eta_2 | c_1\eta_1 + c_2\eta_2 \rangle} = \frac{|c_1|^2 h_{11} + 2c_1c_2 h_{12} + |c_2|^2 h_{22}}{|c_1|^2 + |c_2|^2}$$

Where, as before, $\langle \eta_1 | \hat{h} | \eta_1 \rangle = h_{11}$ and so on. Taking the derivative with respect to c_1 and setting it equal to 0,

$$\frac{\partial E}{\partial c_1} = \frac{2c_1 h_{11} + 2c_2 h_{12}}{|c_1|^2 + |c_2|^2} - \frac{2c_1 |c_2|^2 h_{22}}{(|c_1|^2 + |c_2|^2)^2} = 0$$

Notice that $\frac{|c_2|^2 h_{22}}{|c_1|^2 + |c_2|^2}$ is the expression E for the energy of η_2 , normalized. Solving for E , we get that $Ec_1 = c_1 h_{11} + c_2 h_{12}$. If we perform the same computation with $\frac{\partial E}{\partial c_2}$, then we get

$$\begin{aligned} c_1 h_{11} + c_2 h_{12} &= Ec_1 \\ c_2 h_{22} + c_1 h_{21} &= Ec_2 \end{aligned}$$

Which can be conveniently represented like so

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = E \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Which is the same equation that we had before, but derived in a different way from the same assumptions. It will yield two eigenvalues, one representing a minimum in energy (the one we are looking for with the variational theorem, which ends up being the bonding orbital), and one representing the other point where the derivative is 0, so the energy maximum or the antibonding orbital.

4 Perturbation Theory

4.1 Theory

This part of the course is covered in a lot of detail in the textbook. The fundamental results are that the first order correction E_n^1 is

$$E_n^1 = \langle \psi_n^0 | \hat{H}^1 | \psi_n^0 \rangle$$

where ψ_n^0 is the unperturbed wavefunction and $\hat{H}^1$ is the perturbation Hamiltonian. Similarly,

$$E_n^2 = - \sum_{m \neq n} \frac{\langle \psi_m^0 | \hat{H}^1 | \psi_n^0 \rangle \langle \psi_n^0 | \hat{H}^1 | \psi_m^0 \rangle}{E_m^0 - E_n^0}$$

4.2 Application: Molecule in an Electric Field

Compared to the nuclear electric field, an external electric field (such as one applied during spectroscopy), is only a small perturbation, so can be treated with perturbation theory.

First, let's look at the problem classically. The electric field is just a change in potential, and is defined as $\vec{\epsilon} = -\vec{\nabla}V$. The energy of charges in an electric potential is simply $E = \sum_i q_i V(\vec{r}_i)$, where $\vec{r}_i$ is the position of charge i . We make the assumption that the electric field is constant. While that isn't exactly true during spectroscopy, we are dealing with size scales on the order of molecules, so it is a very good approximation.

$\frac{\partial V}{\partial z} = \epsilon_z$, so $V(\vec{r}) = -\epsilon_z \cdot z$. So in the z direction, the change in energy of charges in the electric field is $\sum_i q_i \epsilon_z z_i$. We define the *dipole moment* as $\mu_z = \sum_i q_i z_i$, and similarly for directions x and y . Then $\Delta E = -\mu_z \epsilon_z$.

Aside: We would like the dipole moment to be independent of origin. This is true only if the molecule is neutral, since for some constant c , $\sum q_i(z_i + c) = \sum q_i z_i$ if $\sum q_i = 0$. When we talk about the dipole moment of a charged molecule in organic chemistry, we are actually referring to an average value.

Now that we've established the classical theory of the dipole moment, we can approach this problem with quantum mechanics. The perturbation causes a change in the potential that is $-\mu_z \epsilon_z$, so $H = H^0 - \hat{\mu}_z \epsilon_z$, for the operator $\hat{\mu}$.

Calculating the first order energy correction, $E_0^1 = \langle \psi_0^0 | \hat{\mu}_z \epsilon_z | \psi_0^0 \rangle = \epsilon_z \langle \psi_0^0 | \hat{\mu}_z | \psi_0^0 \rangle$. Thus, a molecule with no average dipole moment would have no first order change in the energy when in an electric field.

If we calculate the second order correction (and omit the algebra, for now), we get that

$$E = E^0 - \langle \mu \rangle \epsilon + \frac{1}{2} \langle \alpha \rangle \epsilon^2$$

For some value α . This is what we call the *electric polarizability*.

If we think about these energy corrections as a form of *effective dipole* μ_{eff} such that $E = E^0 - \epsilon \mu_{eff}(\epsilon)$, then $\mu_{eff} = \langle \mu \rangle - \frac{1}{2} \langle \alpha \rangle \epsilon$. Thus the induced dipole is proportional to the polarizability.

In some organic chemistry textbooks, this value is described as the “squishiness” of a molecule. Now we know that is true, as we’ve measured how easy it is for a molecule to become polarized in an electric field.

Now, let’s actually compute what this value is.

$$\langle \alpha \rangle = 2E_0^2 = 2 \sum_{n \neq 0} \frac{\langle \psi_0^0 | \mu_z | \psi_n^0 \rangle \langle \psi_n^0 | \mu_z | \psi_0^0 \rangle}{E_n^0 - E_0^0}$$

For the purpose of simplicity, we assume that there exists some sort of “average excitation” Δ such that for any $n \neq 0$, $E_n^0 - E_0^0 = \Delta$. If we sum over all n and then subtract the case where $n = 0$,

$$\langle \alpha \rangle = \frac{2}{\Delta} \sum_n \langle \psi_0^0 | \mu_z | \psi_n^0 \rangle \langle \psi_n^0 | \mu_z | \psi_0^0 \rangle - \langle \psi_0^0 | \mu_z | \psi_0^0 \rangle \langle \psi_0^0 | \mu_z | \psi_0^0 \rangle$$

We use the fact that $\sum_n |\psi_n^0\rangle \langle \psi_n^0| = 1$ (a fact we called *the resolution of the identity* earlier in the course), then

$$\langle \alpha \rangle = \frac{2}{\Delta} (\langle \psi_0^0 | \mu_z^2 | \psi_0^0 \rangle - \langle \psi_0^0 | \mu_z | \psi_0^0 \rangle^2) = \frac{2}{\Delta} \text{Var}(\mu_z)$$

Thus, a large polarizability means we have large fluctuations in the measurement of the dipole.

Example (Spherical Atom). If we consider a spherical atom, $\mu_z = q \cdot z$, then $\langle \alpha \rangle = \frac{2}{\Delta} (\langle z^2 \rangle - \langle z \rangle^2) q^2$

This short example tells us that the polarizability is proportional to the variability in the size of the atom, and inversely proportional to the excitation energy. This makes sense, as in general chemistry we learn that bigger atoms that are easily excited are the most polarizable.

4.3 Application: Intermolecular Interactions

4.3.1 Theory

From general chemistry, we know that molecules can interact with each other via charge-charge, dipole-dipole, or london disperion force (Vanderwaals) interactions, as well as any pairwise combination of these phenomena. The existence of all of these interactions can be derived with perturbation theory.

If we have molecules A and B, then an intermolecular interaction between them can be measured as a change in their overall energy. That is,

$$E_{AB} = E_A = E_B + \Delta_{AB}$$

where Δ is the interaction energy. If $\Delta > 0$, then the force is repulsive (bringing the molecules closer increases the energy) and if $\Delta < 0$, then it is attractive.

Writing the Hamiltonians for the electrons of A, then

$$H_A = T_A + V_{\text{nucleus of A}} + V_{\text{electrons of A}}$$

where T is the kinetic energy of the electrons, $V_{\text{nucleus of A}}$ is the potential on the electrons due to the nuclei of A, and $V_{\text{electrons of A}}$ is the potential on the elctrons from each other. A similar expression can be written for H_B .

When we bring these molecules near each other, then the Hamiltonian can be written

$$H_A + H_B + V_{\text{nuclei-electrons, AB}} + V_{\text{electrons, AB}}$$

where $V_{\text{nuclei-electrons, AB}}$ is the potential on the electrons of A by the nuclei of B, as well as the potential on the electrons of B by the nuclei of A, and $V_{\text{electrons, AB}}$ is the potential of the electrons on A by the electrons on B and vice versa.

We write the 0th Hamiltonian $H^0 = H_A + H_B$. Then, we attempt to write the 1st order Hamiltonian as best we can with the interactions we know may occur.

$$H^1 = \frac{\hat{q}_A \hat{q}_B}{R_{AB}} + \frac{\hat{q}_A \hat{\mu}_B}{R_{AB}^2} + \frac{\hat{q}_B \hat{\mu}_A}{R_{AB}^2} + \frac{\hat{\mu}_A \hat{\mu}_B}{R_{AB}^3} + \dots$$

There may be other terms, but these are the interactions we know that occur so we can use it to calculate the most significant terms of the first order correction to the energy. Note that this expression required assuming that the atoms are far enough from each other that the interactions can be treated as interactions between point atoms.

What are the charge and dipole moment operators? The charge operator measures the total charge of a molecule. Precisely, $\hat{q} = \int \delta(r - r_i) q_i dr$ for each particle i in the molecule. What's important about this operator is that each eigenstate of a molecule is in a definite state of charge. Thus, if we measure the charge of an eigenstate, we will get one value only, and that is the eigenvalue of the charge operator, and thus the charge of the molecule.

The other important thing about the charge operator is that, by nature of returning a specific value when measured, $\langle \psi_{0A}^0 | \hat{q}_A | \psi_{0A}^m \rangle = \langle \psi_{0A}^0 | q_{Am} | \psi_{0A}^m \rangle$, where q_{Am} is the charge on A in state m . Since ψ_{0A}^0 is orthogonal to ψ_{0A}^m , then

$$\langle \psi_{0A}^0 | \hat{q}_A | \psi_{0A}^m \rangle = q_{Am} \langle \psi_{0A}^0 | \psi_{0A}^m \rangle = 0$$

.

Similarly, the dipole moment operator $\hat{\mu}$ measures the dipole moment of a molecule. Since $\hat{m}u = \sum_i r_i q_i$, an eigenstate of a molecule is not necessarily in a definite state of the dipole moment operator. Thus, $\langle \psi_0^0 | \hat{\mu} | \psi_0^m \rangle \neq 0$.

4.3.2 The Vanderwaals Force

Let us now consider the interaction energy between two neutral, spherical atoms A and B. For the sake of clarity, we will write $\psi_{0A}^0 = 0_A$ for the ground state of A (and a similar expression for B), $\psi_{0A}^m = m_A$ for an excited state of A, and $\psi_{0B}^n = n_B$ for an excited state of B.

Also, we write the overall wavefunction as the product of wavefunctions of the constituent parts, that is $\psi_0^0 = 0_A 0_B$. Finally, since the atoms are neutral and spherical, we know $\langle 0_A | \hat{q}_A | 0_A \rangle = 0$ and $\langle 0_A | \hat{\mu}_A | 0_A \rangle = 0$, and identically for B.

The first order correction of the energy is

$$E^1 = \langle 0_A 0_B | H^1 | 0_A 0_B \rangle$$

We simply need to evaluate the terms of H^1 one by one. For example,

$$\langle 0_A 0_B | \frac{\hat{\mu}_A \hat{\mu}_B}{R_{AB}^3} | 0_A 0_B \rangle = \frac{1}{R_{AB}^3} \langle 0_A | \hat{\mu}_A | 0_A \rangle \langle 0_B | \hat{\mu}_B | 0_B \rangle = 0$$

Since $\hat{\mu}_A$ is only dependent on A and $\hat{\mu}_B$ is only dependent on B, we can split the calculation into two, and we know that both those terms are 0.

Skipping the algebra for the rest of the terms (the problem is identical, except with charges), overall $E^1 = 0$. Thus, to first order, two spherical, neutral atoms don't interact.

Again,

$$E^{(2)} = - \sum_{m \neq 0} \frac{\langle \psi_0^0 | H^1 | \psi_m^0 \rangle \langle \psi_m^0 | H^1 | \psi_0^0 \rangle}{E_m^0 - E_0^0}$$

Once again, we let Δ be some average excitation value that takes into account the denominator for all $m \neq 0$. We calculate the expectation values for the terms of H^1 one by one. For example, consider the term $\frac{\hat{q}_A \hat{q}_B}{R_{AB}}$:

$$\langle 0_A 0_B | \hat{q}_A \hat{q}_B | m_A m_B \rangle = \langle 0_A | \hat{q}_A | m_A \rangle \langle 0_B | \hat{q}_B | n_B \rangle$$

Without loss of generality, if $m = 0$, then the term is $\langle 0_A | \hat{q}_A | 0_A \rangle$, which is 0 as the atom is neutral. If $m \neq 0$, then the atom A is still in a definite state of the charge operator, and thus for the charge of the atom q in the state m , $\langle 0_A | \hat{q}_A | m_A \rangle = q \langle 0_A | m_A \rangle = 0$ since these two states are orthogonal. Thus, this term of the Hamiltonian evaluates to 0.

Similarly, the $\frac{\hat{q}_A \hat{\mu}_B}{R_{AB}^2}$ and $\frac{\hat{q}_B \hat{\mu}_A}{R_{AB}^2}$ terms will evaluate to 0.

The one remaining term is $\frac{\hat{\mu}_A \hat{\mu}_B}{R_{AB}^3}$. Thus,

$$E^{(2)} = - \sum_{m \neq 0} \frac{\langle 0_A 0_B | \hat{\mu}_A \hat{\mu}_B | m_A n_B \rangle \langle m_A n_B | \hat{\mu}_A \hat{\mu}_B | 0_A 0_B \rangle}{R_{AB}^6 \Delta}$$

$\langle 0_A 0_B | \hat{\mu}_A \hat{\mu}_B | m_A n_B \rangle = \langle 0_A | \hat{\mu}_A | m_A \rangle \langle 0_B | \hat{\mu}_B | n_B \rangle$, which is only non-zero if both m and n are not the ground state. Thus, *both* atoms have to be in an excited state for this term to not be 0.

$$E^{(2)} = - \frac{1}{R_{AB}^6 \Delta} \sum_{m, n \neq 0} (\langle 0_A | \hat{\mu}_A | m_A \rangle \langle m_A | \hat{\mu}_A | 0_A \rangle) (\langle 0_B | \hat{\mu}_B | n_B \rangle \langle n_B | \hat{\mu}_B | 0_B \rangle)$$

The sum is not 0, but more precisely,

$$\begin{aligned} \sum_{m, n \neq 0} \langle 0_A | \hat{\mu}_A | m_A \rangle \langle m_A | \hat{\mu}_A | 0_A \rangle &= \alpha_A \\ \sum_{m, n \neq 0} \langle 0_B | \hat{\mu}_B | n_B \rangle \langle n_B | \hat{\mu}_B | 0_B \rangle &= \alpha_B \end{aligned}$$

So $E^{(2)} = - \frac{1}{R_{AB}^6 \Delta} \alpha_A \alpha_B$. This is an attractive force caused by dipoles inducing dipoles on each other, and is proportional to the polarizability of each atom. This is the source of the Vanderwaals or London Dispersion Force.